

Excel Cleaning Report — Retail Sales Data (Jan 2023 - Dec 2024)

Generated on: 2025-10-07 10:40:18

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Dataset: Retail Sales (1500 rows; Orders between 2023-01-01 and 2024-12-31)

Background of Study

Companies collect transactional sales data from multiple sources (POS, e-commerce, ERPs). Such raw data often contains inconsistencies, missing values, mixed formats and noise. Before performing any analysis—trend detection, dashboarding or modeling—the raw data must be cleaned and validated. This document describes the Excel-based cleaning process applied to a sample retail sales dataset covering January 1, 2023 through December 31, 2024 (1500 rows). The goal was to transform messy, real-world data into an analysis-ready dataset while documenting every change for reproducibility and client trust.

Aim of Study

The primary aim of this cleaning project was to convert the raw retail sales dataset into a clean, consistent and validated Excel file suitable for analysis and dashboarding. Specific objectives include:

- - Identify and document data quality issues (missing data, inconsistent formats, duplicates, outliers).
- - Apply conservative and explainable imputation methods where necessary, while retaining provenance (imputation flags).
- - Standardize categorical values and data types for accurate aggregation and joins.
- - Create derived features to support downstream analysis and Power BI dashboarding.

About the Dataset

Key facts about the dataset:

- - Period: 1 January 2023 to 31 December 2024
- - Rows: 1500 (transaction-level / line items)
- - Typical columns: Order_ID, Order_Date, Ship_Date, Ship_Mode, Customer_ID, Customer_Name, Segment, Country, City, State, Postal_Code, Region, Product_ID, Category, Sub_Category, Product_Name, Sales, Quantity, Discount, Profit

The dataset originally contained the following missing-value counts identified during profiling:

Column	Missing Count (initial)
Customer_ID	30
Product_ID	34
Sales	11
Quantity	15
Discount	18
Profit	12
Ship_Date	26

Tools and Techniques Used

Primary tools and Excel features used in this cleaning project:

- - Microsoft Excel (desktop) — formulas, formatting, and tables.
- - Power Query (Get & Transform) — reproducible import, type conversion, trimming and step logging.
- - Key Excel formulas: TRIM, CLEAN, SUBSTITUTE, MEDIAN, IF, ISBLANK, COUNTBLANK, TEXT, DATEVALUE, VALUE.
- - Excel features: Ctrl+G (Go To Special → Blanks), Conditional Formatting, Remove Duplicates, PivotTables, Text to Columns, Paste Special, Format Cells, Data Validation.
- - Documentation: change log sheet, imputation flag columns, data dictionary (columns + description).

Step-by-Step Cleaning Process (Detailed)

1. Backup Original File

What: Save an untouched copy of the raw dataset before any modification.

Why: Prevents accidental data loss and provides a rollback point.

How (Excel): File → Save As → Retail_Sales_Raw_backup.xlsx. Work on a copy named Retail_Sales_Work.xlsx.

Benefit: Ensures reproducibility and auditability.

2. Inspect (shape, head, missing %)

What: Quick data profiling to understand size, sample records, data types and missingness.

Why: Guides priorities for cleaning and sets expectations.

How (Excel): Open sheet, use COUNTA/ROWS/COLUMNS or Excel Status Bar for count. Use conditional formatting and COUNTBLANK(range) per column. Use PivotTables to inspect categories and ranges.

Benefit: Identifies problem columns early and estimates effort required.

3. Standardize Column Names

What: Rename columns consistently (snake_case or Title Case).

Why: Clean names improve readability and avoid errors during automation, imports and scripting.

How (Excel): Edit header row cells or use Power Query: Transform → Use First Row as Headers → Rename columns: order_id, order_date, ship_date, etc.

Benefit: Consistent column naming reduces mistakes when writing formulas, DAX or code.

4. Trim & Clean Text Columns

What: Remove leading/trailing spaces and hidden non-printable characters from text fields.

Why: Invisible characters cause false duplicates and grouping errors (e.g., 'Claire ' ≠ 'Claire').

How (Excel formulas):

- - Helper column formula: =TRIM(CLEAN(SUBSTITUTE(E2,CHAR(160),"")))
- - Fill down and Paste Special → Values back into original column.
- - Alternatively use Power Query: Transform → Format → Trim and Clean.

Benefit: Ensures exact matches for joins and grouping; fixes Ctrl+G blank-detection issues.

5. Parse Dates & Convert Numerics

What: Convert date-like columns into Excel Date type and numeric columns into numbers.

Why: Proper types are required for arithmetic, sorting, date functions, and charts.

How (Excel):

- - For mixed date formats: =DATEVALUE(SUBSTITUTE(A2,"-","/")) or use Power Query to change type using locale.
- - For numerics stored as text: use VALUE(SUBSTITUTE(...)) or Text to Columns trick.

Benefit: Enables accurate calculations (ship lag, month grouping, sums, averages).

6. Flag Missing Values and Create Imputed Flag Columns

What: For every column where imputation is possible, create a flag column that records if the value was originally missing.

Why: Flags preserve provenance and transparency for clients and future reviewers.

How (Excel):

- - Add columns e.g., sales_missing_flag with formula: =IF(ISBLANK(Q2),1,0)
- - Use COUNTIF on flags to log imputed counts into ChangeLog.

Benefit: Maintains traceability and allows filtering to inspect imputed rows.

7. Detect Duplicates and Mark (Don't Delete Immediately)

What: Identify exact duplicates and duplicates by business keys (e.g., order_id + product_id + order_date).

Why: Duplicates can distort aggregates; but some repeated order_ids can be legitimate (line items), so do not delete without inspection.

How (Excel):

- - Use Conditional Formatting → Duplicate Values for quick scan.
- - Create a helper key column: =order_id & "|" & product_id & "|" & TEXT(order_date,"yyyy-mm-dd") and use COUNTIF(key_range,key_cell).
- - Create duplicate_flag = IF(COUNTIF(key_range,key_cell)>1,1,0).

Benefit: Ensures correct totals and removes accidental double-records only after manual review.

8. Standardize Categories

What: Normalize categorical column values (Ship_Mode, Category, Sub_Category, State, Postal_Code formatting).

Why: Different spellings or formats split groups and misrepresent group totals.

How (Excel):

- - Use Find & Replace to correct common variants (e.g., 'Std' -> 'Standard Class').
- - Postal code correction: use =TEXT(L2,"000000") for 6-digit standardization or manual correction based on city/state.

Benefit: Ensures accurate grouping and that maps/pivots aggregate correctly.

9. Validate Business Rules

What: Validate logical constraints such as Ship_Date >= Order_Date and Sales ≥ 0.

Why: Ensures records reflect real-world processes and detects entry errors.

How (Excel):

- - Add helper: `ship_lag_days = [@Ship_Date]-[@Order_Date]`, then filter `ship_lag_days < 0`.
- - Use formulas: `=IF(OR(Sales<=0,Quantity<=0),"Check","OK")`.

Benefit: Catches impossible or suspicious records for correction before analysis.

10. Handle Missing Values and Impute Conservatively

What: Apply conservative, explainable imputation rules and record them.

Why: Preserves dataset size for analysis while maintaining transparency.

How (Imputation rules used in this project):

- - Customer_ID: fill with 'Unknown'
- - Product_ID: fill with 'MissingProduct'
- - Discount: fill with 0
- - Sales: fill with `MEDIAN(sales_range)` using `=MEDIAN(Q2:Q1501)`
- - Quantity: fill with `ROUND(MEDIAN(quantity_range),0)`
- - Profit: compute median profit margin = `MEDIAN(IF(sales<>0,profit/sales))`, then profit = median_margin * sales for missing profits
- - Ship_Date: impute as Order_Date + median shipping lag (calculated per ship_mode if available), and flag `ship_date_imputed_flag`

Excel mechanics:

- - Use `Ctrl+G` → Special → Blanks to select blanks and paste formulas like `=$AE$1 (median)` then Paste Special → Values.
- - Create imputation flag columns before filling: e.g., `sales_imputed_flag = IF(ISBLANK(Q2),1,0)`.

Benefit: Imputation provides plausible values while preserving ability to filter and audit imputed rows.

11. Detect & Flag Outliers

What: Identify profit outliers using the IQR method and flag them for review.

Why: Outliers can distort averages and model training.

How (Excel):

- - Calculate Q1 and Q3: `=QUARTILE.INC(T2:T1501,1)` and `=QUARTILE.INC(T2:T1501,3)`
- - IQR = Q3 - Q1. Bounds: Lower = `Q1 - 1.5*IQR`, Upper = `Q3 + 1.5*IQR`
- - Flag: `=IF(OR(T2<Lower,T2>Upper),"Outlier","Normal")`.

Benefit: Enables focused inspection and informed decisions (cap, remove or retain) on extreme profit values.

12. Create Derived Columns (Feature Engineering)

What: Add new columns that make analysis easier and richer.

Why: Derived features support segmentation, trend analysis and advanced visuals.

How (Excel formulas examples):

- - order_year: =YEAR(order_date)
- - order_month: =TEXT(order_date,"yyyy-mm")
- - processing_days: =ship_date - order_date
- - profit_margin: =IF(sales=0,NA(),profit/sales)
- - sales_per_unit: =sales/quantity
- - discount_band: =IF(discount=0,"0%",IF(discount<=0.1,"1-10%",IF(discount<=0.2,"11-20%","21%+")))
- - high_value_flag: =IF(sales>5000,"High","Regular")

Benefit: Feature columns speed up pivoting, filtering and plotting; they are inputs to dashboards and models.

13. Reconcile Totals and Sanity-Check Results

What: Compare sums and counts before and after cleaning and ensure consistency.

Why: Confirms no accidental row deletion and verifies aggregates are within expected ranges.

How (Excel):

- - Compare SUM(Sales) before vs after and verify Orders count = 1500.
- - Use PivotTables to compare totals by Month/Region and spot unexpected changes.

Benefit: Validates cleaning work and builds confidence in downstream analysis.

14. Save Cleaned File and Document Steps

What: Save final cleaned workbook and record all changes in a ChangeLog sheet.

Why: Ensures reproducibility, audit trail and client-ready deliverable.

How (Excel):

- - Save as Retail_Sales_Clean_v1.xlsx and include a sheet named ChangeLog with columns Date, Step, Column, MissingBefore, Method, ImputedCount, Notes.
- - Keep flags and derived columns in the file for transparency.

Benefit: Final deliverable is ready for Power BI dashboarding and modeling; client can review the change log.

Change Log (Sample Entries)

Date	Step	Column	Missing Before	Method	Notes
2025-10-07	Impute	Customer_ID	30	Fill 'Unknown'	Flag: customer_id_imputed_flag
2025-10-07	Impute	Product_ID	34	Fill 'MissingProduct'	Flag: product_id_imputed_flag
2025-10-07	Impute	Sales	11	Fill median	Flag: sales_imputed_flag
2025-10-07	Impute	Ship_Date	26	OrderDate + median lag by ship_mode	Flag: ship_date_imputed_flag

Appendix: Key Excel Formulas & Shortcuts

Common formulas used in this project:

Trim + Clean + Substitute (remove non-breaking spaces):

=TRIM(CLEAN(SUBSTITUTE(A2,CHAR(160),"")))

Median (ignore blanks): =MEDIAN(Q2:Q1501)

COUNTBLANK: =COUNTBLANK(range)

IQR bounds: Q1 = QUARTILE.INC(range,1); Q3 = QUARTILE.INC(range,3); IQR = Q3-Q1

Impute with median (using blanks selection): Select blanks → type =\$AE\$1 → Ctrl+Enter → Paste Special → Values

Profit margin: =IF(Sales=0,NA(), Profit/Sales)

Useful shortcuts: Ctrl+G → Special → Blanks; Ctrl+Enter (fill selected blanks); Alt+H+V+S (Paste Special).

Next Steps & Recommendations

- Use this cleaned dataset to build the Power BI dashboard (dashboard guide created separately).
- Retain imputation flags and change log in the deliverable so the client can review all changes.
- If the dataset will be refreshed regularly, move reproducible steps into Power Query and save the query steps for scheduled refresh.

- For statistical modeling, consider additional feature engineering and handling profit outliers with domain-informed corrections.